dynamic planet C

SOLVI 2022



Names:___KEY_____

Team Name: _____ Team Number: _____

Score: _____ / 260 Rank: _____

Brief Instructions

- **Section 1 (Multiple choice and short answer questions)** begins on page 2.
- **Section 2 (Calculations)** begins on page 9.
- There are 7 tiebreakers, denoted by **{TB#}**: #56, #23, #43, #47, #3, #22, #63. If needed, score on Section 2 will serve as an 8th tiebreaker.
- If you have any questions/comments/feedback about this exam after the competition, feel free to reach out to either of us through our listed emails. Good luck and have fun!

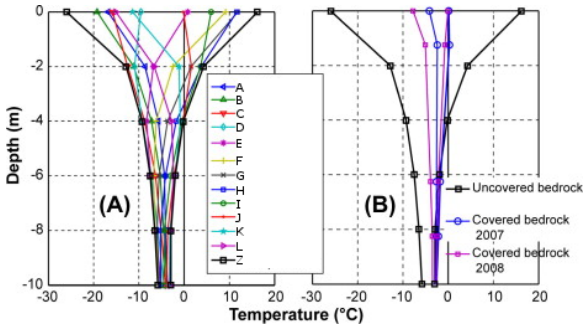
Test Writers

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Section 1: MCQ and SAQ Mashup! [165 pts]

For questions 1-55, answer to the best of your ability! You got this :) Each question in this section is worth 3 points.

Use Diagram A to answer the next two questions.



1. What is the name of the curve shown?
    a. Bell curve
    b. Trumpet curve
    c. Funnel curve
    d. Double-logistic curve
2. Which of the following is a realistic association between each curve and a date of measurement? These measurements were taken in the northern hemisphere.
    a. A Jan '08 // D Apr '08 // H Aug '08 // L Jan '09
    b. A Jan '08 // D Mar '08 // H May '08 // L Aug '08
    c. A Jun '08 // D Oct '08 // H Apr '09 // L Jun '09
    d. A Jun '08 // D Aug '08 // H Nov '08 // L Jan '09
3. Which of the following velocities would allow for the transport of the smallest pebble? **{TB5}**
    a. 5 cm/s
    b. 20 cm/s
    c. 100 cm/s
    d. 500 cm/s
4. Which of the following is a characteristic of mountainous watersheds?
    a. Low soil porosity
    b. High infiltration
    c. Low runoff
    d. High saltwater intrusion
5. If a stream's velocity is 100 centimeters per second, then what are the largest particles that this stream can transport?
    a. Sand
    b. Boulder
    c. Cobble
    d. Silt
6. At minimum, what length of time must the ground remain completely frozen in order to be classified as permafrost?
    a. 1 year
    b. 2 years
    c. 4 years
    d. 8 years
7. Which of the following is a characteristic of oligotrophic lakes?
    a. High turbidity
    b. Many small fish present
    c. High nutrient levels
    d. Stratification during summer
8. Which of the following is NOT a good aquifer?
    a. Granite
    b. Gravel
    c. Limestone
    d. Sandstone

9. What two processes are most likely to lead to a rise in the water table?
    a. Infiltration and precipitation
    b. Infiltration and evaporation
    c. Erosion and precipitation
    d. Erosion and evaporation
10. In a meandering stream, stream erosion is highest _____ and stream velocity is highest _____.
    a. Inside of the curve; inside of the curve
    b. Inside of the curve; outside of the curve
    c. Outside of the curve; inside of the curve
    d. Outside of the curve; outside of the curve

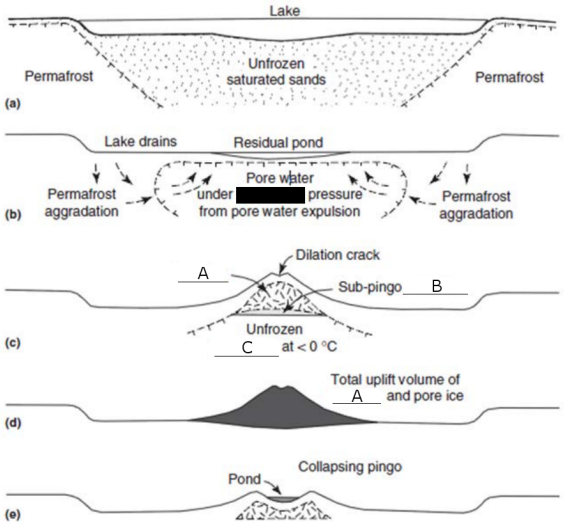
The next three questions will involve topographic maps. Please reference the attached maps.

11. Reference section 27 of township T20S R14E of the Ottumwa quadrangle from 2018. Select any and all water features present in this section according to the map. (Note: not all features present may be available answer choices)
    a. Marsh or swamp
    b. Submerged marsh or swamp
    c. Wooded marsh or swamp
    d. Intermittent river
    e. Intermittent stream
    f. Perennial lake or reservoir
12. Now reference the same location but on the 1970 map. Once again, select any and all water features present in this section according to the map. (Note: not all features present may be available answer choices)
    a. Marsh or swamp
    b. Submerged marsh or swamp
    c. Wooded marsh or swamp
    d. Intermittent river
    e. Intermittent stream
    f. Perennial lake or reservoir
13. Finally, reference the 1894 map of Burlington in conjunction with the previous two maps. The Ottumwa quadrangle is approximately center-left on this map. What change may have taken place between 1894 and 1970 to cause such a dramatic change in the landscape?

    The John Redmond Dam was constructed along the Neosho River, creating the John Redmond Reservoir. (This is noted at the bottom of the 1970 map)

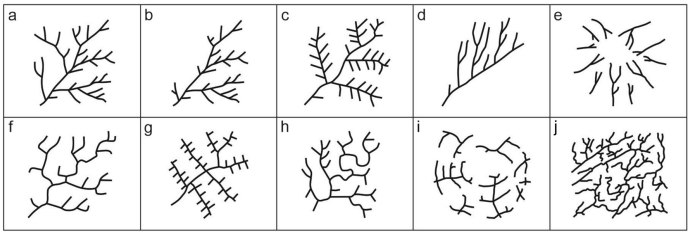
14. On a topographic map, the point of the "V" shapes in contour lines around rivers and streams point _____.
    a. Upstream
    b. Downstream
    c. Toward the banks
    d. Toward the center

Use this image to answer the next two questions.



15. Identify each component of pingo formation for each letter.
    a. Pingo ice (1-ice)
    b. Water lens (½-water; ½-lens)
    c. Pore water (½-pore; ½-water)
16. Which term(s) may accurately describe this pingo?
    a. Open-system
    b. Closed-system
    c. Hydraulic
    d. Hydrostatic

Use this image to answer the next three questions. Each Box is labeled with a letter and displays a different drainage pattern.



17. Identify the drainage pattern depicted in Box (c).
    a. Annular
    b. Pinnate
    c. Trellis
    d. Parallel
18. Which Box depicts a drainage pattern that can develop in an area where sedimentary rocks have been folded and then eroded?
    a. (a)
    b. (d)
    c. (g)
    d. (j)
19. Which Box depicts a drainage pattern that is often exhibited in areas with volcanoes?
    a. (b)
    b. (e)
    c. (f)
    d. (i)
20. What type of waterfall falls vertically without making contact with the underlying face of the cliff?
    a. Plunge
    b. Horsetail
    c. Fan
    d. Segmented

21. The International Waterfall Classification System utilizes the Beisel rating to classify waterfalls. This rating is calculated by taking the natural logarithm of what quantity that is characteristic of waterfalls?
    a. Time it takes for water to reach the pool
    b. Height of the fall
    c. Average discharge of the watercourse
    d. Volume of water present
22. Below is an image that depicts what type of waterfall? **{TB6}**



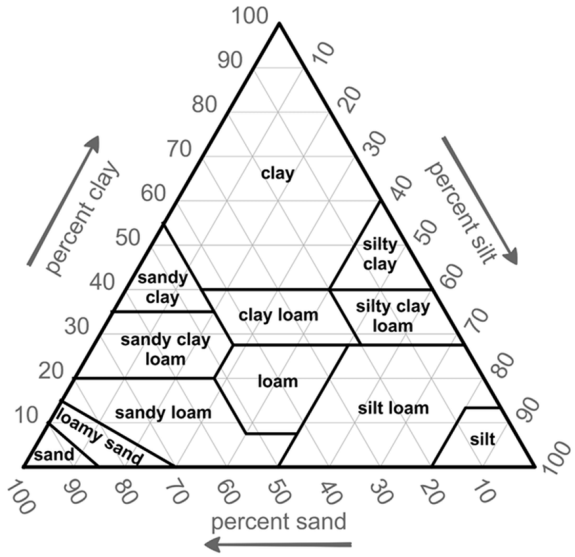
Tiered fall

23. Which lake zone is sometimes referred to as the "lake's digestive system," as bacteria decompose organic matter in this area? **{TB2}**
    a. Euphotic zone
    b. Benthic zone
    c. Photic zone
    d. Profundal zone
24. Which lake zone often receives an abundance of nutrients from nonpoint source pollution due to it being shallow?
    a. Littoral zone
    b. Profundal zone
    c. Benthic zone
    d. Aphotic zone
25. What separates the epilimnion from the hypolimnion?

    Accept either thermocline *or* metalimnion

26. Which term describes the average time that it takes to completely renew the water volume of a lake?
    a. Intermittent time
    b. Flushing time
    c. Polymictic time
    d. Recharge time
27. Which term is used to describe the lateral movement of water in the vadose zone?
    a. Overland flow
    b. Streamflow
    c. Interflow
    d. Base flow
28. Which of the following are typically **not** found in thermokarst regions?
    a. Lakes and ponds
    b. Hummocks
    c. Limestone columns
    d. Pingos

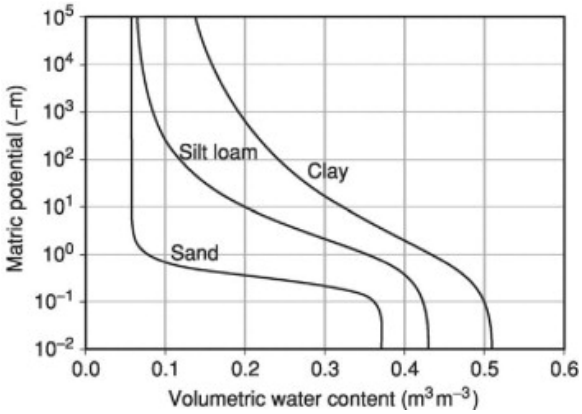
Use the following soil triangle to answer the next two questions.



29. What type of soil consists of 25% silt and 40% sand?
    a. Sandy clay
    b. Clay
    c. Clay loam
    d. Silty clay
30. Of the following soil types, which has the lowest hydraulic conductivity?
    a. Silty clay
    b. Loam
    c. Loamy sand
    d. Silt loam
31. Which of the following are dominated by trees that are water tolerant?
    a. Bogs
    b. Marshes
    c. Swamps
    d. None of the above
32. Which of the following are typically acidic?
    a. Bogs
    b. Marshes
    c. Swamps
    d. None of the above
33. Which of the following is characterized by herbaceous vegetation?
    a. Bogs
    b. Marshes
    c. Swamps
    d. None of the above
34. Which of the following are found north and south of the tropics?
    a. Bogs
    b. Marshes
    c. Swamps
    d. None of the above
35. What is the source of most nitrogen pollution in surface water?
    a. Fertilizer
    b. Detergents
    c. Sewage
    d. Rainwater
36. What is the drop in water surface elevation divided by the distance that the water flows?
    a. Hydrologic resistance
    b. Stream gradient
    c. Hydraulic conductivity
    d. Stream discharge
37. Which of the following characteristics are found in all good aquifers?
    a. High porosity, low permeability
    b. Low porosity, high permeability
    c. Low porosity, low permeability
    d. High porosity, high permeability
38. Which of the following characteristics is found in confining beds?
    a. Low hydraulic conductivity
    b. High hydraulic conductivity
    c. Low volumetric soil moisture content
    d. High volumetric soil moisture content

39. What is the term for a fissure made of limestone that is enlarged due to dissolution from rainwater?
    a. Dry sink
    b. Esker
    c. Grike
    d. Moraine
40. For water to be classified as freshwater, it must contain less than _____ mg/L of dissolved solids.
    a. 10
    b. 100
    c. 1,000
    d. 10,000

Use this image to answer the next three questions.



41. What is the name of the curve shown?
    a. Rating curve
    b. Water retention curve
    c. Flow duration curve
    d. Crop coefficient curve
42. At a volumetric water content of 0.3, which of the three soil types shown has the greatest matric potential?

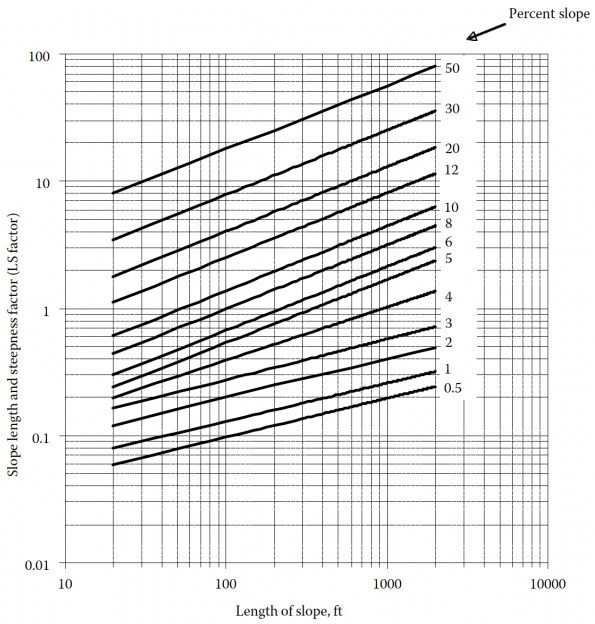
    Sand

43. Calculate the gravimetric soil moisture content of silt loam when it has a matric potential of -1. Assume that the bulk density of silt loam is 1.33 g/cm^3 and the density of water is 1 g/cm^3, and round to two decimal places. **{TB3}**

    $G = (\theta^* \rho_w)/(\rho_b) = (0.35*1)/(1.33) =$ 0.26 g/g [answer can also be expressed without units] *Accept answers between* 0.24 *and* 0.28, *as values for* θ *may be between 0.33 and 0.37*

44. Approximately how much of the extracted groundwater on Earth is used for agricultural purposes?
    a. 5%
    b. 30%
    c. 50%
    d. 70%
45. A dumb farmer poured aresnic into a stream. This stream transports this as the…
    a. Suspended load
    b. Dissolved load
    c. Intrusion load
    d. Bed load

Use this image to answer the next two questions.



46. The LS factor is used in which of the following calculations?
    a. Water balance equation
    b. Darcy's Law
    c. Universal soil loss equation
    d. Chevy and Manning equations
47. Determine the LS factor for a field with a slope steepness of 30% and a slope length of 200 ft. **{TB4}**

    Answers between 10-14

48. Which of the following rocks is best for cave formation?
    a. Limestone
    b. Granite
    c. Sandstone
    d. Marble
49. Which term refers to a dry lake bed?
    a. Eddy
    b. Thalweg
    c. Playa
    d. Arroyo
50. What are the 2 most common nutrients that lead to eutrophication?

    Nitrogen and phosphorus

51. What is the main effect that results from excess nutrients in rivers, lakes, and oceans?
    a. Algae growth decreases
    b. Algae growth increases
    c. Algae growth remains constant
    d. Algae is unable to grow
52. What type of flood causes the most deaths?
    a. Coastal
    b. River
    c. Wetland
    d. Flash
53. What is the name of an abandoned, cutoff, meander loop?

    Accept Oxbow lake *or* Oxbow

54. Which of these factors does not determine the rate of stream erosion?
    a. Stream shape
    b. Stream slope
    c. Stream temperature
    d. Stream velocity
55. What is the artificial reservoir of water behind a dam called?

    Forebay

Section 2: Computation Conundrums! [95 pts]

For questions 56-66, you MUST show all work and include units! Partial credit will be available for almost every question, and an answer without work will NOT be able to earn full credit. Point values are listed next to each question.

56. The evaporation from Lake Pikachu is to be calculated using the water budget method. Inflow to the lake is mainly through two small rivers: A and B. The outflow is through river C. Calculate the evaporation from the lake surface during summer (May - August) if the water level was at +482.32 (m) on May 1 and +482.41 (m) on August 31. The surface area of the lake is 80 km^2, and the precipitation during the period was 800 mm. The average flows in the rivers are given below. (Hint: Be careful when choosing units in the water balance!) [11 pts] **{TB1}**

River	Catchment (km^2)	Q, average (m^3/s)
A	130	23.0
B	105	18.0
C		38.0

Storage, $S = (Q_{in} + P) - (Q_{out} + E)$; P is precipitation, E is evaporation, and Q is flow *[1 pt for the water balance equation]*
May to August = (31 + 30 + 31 + 31) = 123 days *[1 pt]*
$Q_{in} = (23.0 + 18.0\ m^3/s) \times (123 \times 24 \times 3600\ s)/(80 \times 10^6\ m^2) = 5.446$ m *[2 pt]*
P = 0.800 m *[1 pt]*
$Q_{out} = (38.0\ m^3/s) \times (123 \times 24 \times 3600\ s)/(80 \times 10^6\ m^2) = 5.048$ m [2 pt]
$\Delta S = 482.41 - 482.32 = 0.090$ m *[1 pt]*
Now, using all of these calculated numbers into the water balance equation:

$0.090 = (5.446 + 0.800) - (5.048 + E)$
E = 1.108 m ; Evaporation during this period was 1.108 m, or 1,108 mm. *[3 pt; -1 pt if incorrect/missing units]*

This answer can also be given in terms of volume (i.e. m^3), which would mean doing the water balance equation in terms of m^3 rather than m. However, meters are often used for simplicity, as it makes calculations quicker.

57. Determine the flow velocity and discharge in the clean, straight, full stage stream when the mean flow depth is 4 ft. This stream is low gradient. The bed slope is 0.005, the mean width of the flow is 60 ft, and you can approximate that the banks are vertical. Use a Manning's roughness coefficient of 0.025. [10 pt]

Cross-sectional area, $a = (4\text{ ft})(60\text{ ft}) = 240\text{ ft}^2$ *[1 pt]*
Wetted perimeter, $P = (60 + 4 + 4) = 68\text{ ft}$ *[1 pt]*
Hydraulic radius, $R = a/P = (240\text{ ft}^2 / 68\text{ ft}) = 3.53\text{ ft}$ *[2 pt]*
Flow velocity:

$v = (1.5 / n)R^{2/3}S^{1/2} = (1.5 / 0.025)(3.53)^{⅔}(0.005)^{½} =$ 9.83 ft/s *[3 pt; -1 pt if incorrect/missing units]*

Discharge:

$q = va = (9.83)(240) =$ 2,539.2 ft^3/s *[3 pt; -1 pt if incorrect/missing units]*

**Note: if competitors incorrectly calculate the value of R and/or a, they can earn points if they correctly apply the number in the flow velocity and discharge equations.

58. The following table shows Lake Squirtle's monthly inflows and outflows in cubic meters.

Month	Inflow	Outflow
January	4.5	1.1
February	5.2	3.5
March	6.1	5.2
April	6.9	12.4
May	9.4	5.6
June	2.5	8.2
July	5.1	9.0
August	4.7	4.2
September	5.6	5.1
October	1.3	5.2
November	6.3	1.8
December	7.9	10.4

a. If Lake Squirtle began the year with a storage of 61.7 m^3, what was the storage at the end of the year? [6 pt]
$S = S_{initial} + (\Sigma inflows - \Sigma outflows)$
$S = 61.7 + (-6.2) =$ 55.5 m^3 *[6 pt; -2 pt if incorrect/missing units]*

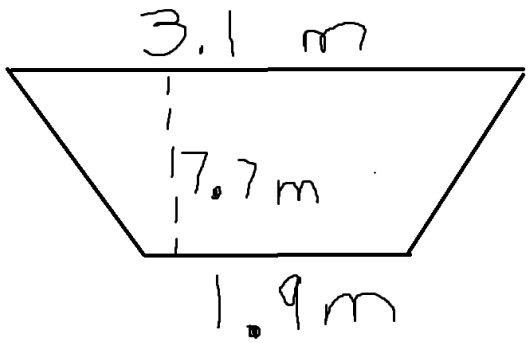
b. In which month was storage the highest? [3 pt]
March

c. In which month was storage the lowest? [3 pt]
October

59. The Darcy flux of a specific water flow region is 5.2 m/min. The porosity of the material through which the water flows is 41%. Find the fluid velocity. Round to the nearest hundredth. [6 pt]

$v = q/n_e$ *[2 pt]*
$v = (5.2)/(0.41) =$ 12.68 m/min *[4 pt; -1 pt if incorrect/missing units, -1 pt if not rounded to the nearest hundredth]*

60. The Charmander Aquifer is 135.4 m long. In the Charmander Aquifer, the soil type is mainly sandy loam. As you may know, the saturated hydraulic conductivity of sandy loam is 2.59 cm/hr. Also, the cross-section of the Charmander Aquifer is trapezoidal, pictured below. Calculate the discharge from this Aquifer if the water table drops 16.3 m. Round to nearest thousandth. [10 pt]



Q = -KiA (Darcy's Law) *[1 pt]*
K = 0.0259 m/hr (or any other equivalent number in m/s, m/min, etc) *[1 pt]*
i = dh/dx = (-16.3 m)/(135.4 m) = -0.12 *[2 pt]*
A = ½(b_1 + b_2)(h) = ½(3.1 + 1.9)(7.7) = 19.25 m^2 *[2 pt]*
Q = -KiA = -(0.029 m/hr)(-0.12)(19.25 m^2) = 0.067 m^3/hr *[4 pt; -1 pt if incorrect/missing units]*

61. The Mew River runs from Pallet Town to Pewter City, and its length is 1683 meters. The straight line distance between Pallet Town and Pewter City is 759 meters.
    a. Calculate the sinuosity of the Mew River. Round to the nearest tenth. [4 pt]
    Sinuosity = 1683 m / 759 m = 2.2 *[4 pt; -2 pt if units are present (sinuosity is a unitless measure]*

    b. Is this river straight or meandering? How do you know? [2 pt]
    Meandering [1 pt], a high sinuosity means meandering [1 pt]

62. Professor Oak has too much time on his hands. One day, he collects 20.4 liters of sand and pours water in it. He notes that the sand absorbs 695 milliliters of water. What is the porosity of this sample? Your answer should be a percentage, and round to the nearest hundredth. [4 pt]
Porosity = 0.695 L / 20.4 L = 3.41% [4 pt, no partial points for this one]

63. Lake Pikachu has two streams connected to it. The Pichu Stream flows into the lake and the Raichu Stream flows out of the lake. The area of Lake Pikachu is 20.0 km^2. Pichu Stream discharges 12.8 m^3/s into the lake, while the Raichu Stream drains 15.9 m^3/s from the lake. Calculate and describe how the level of Lake Pikachu has changed after one year. Round to the nearest thousandth. [10 pt] **{TB7}**

Net flow = inflow - outflow = 12.8 - 15.9 = -3.1 m^3/s *[2 pt]* This means that 3.1 m^3/s are being drained from the lake.
Change in storage after one year: (3.1 m^3/s) x (86400 s / 1 day) x (365 days / 1 yr) x (1 yr) = 97,761,600 m^3 *[2 pt]*
Area of lake: 20.0 km^2 x (1x10^6 m^2 / 1 km^2) = 20 x 10^6 m^2 *[1 pt]*
Change in storage = Area x Change in level
Change in level = (97,761,600 m^3) / (20 x 10^6 m^2) = 4.888 m *[3 pt; -1 pt if incorrect/missing unit]*
The lake will fall 4.888 m *[2 pt for recognizing that the lake's level will decrease]*

64. The Snorlax River has a width of 7.3 meters and an average depth of 2.6 meters. What is the discharge of this river if the flow velocity is 0.04 kilometers per second? [6 pt]

Velocity, v = (0.04 km/s) x (1000 m/km) = 40 m/s *[1 pt]*
Area of river, a = 7.3 x 2.6 = 18.98 m^2 *[2 pt]*
q = va = (40 m/s) x (18.98 m^2) = 759.2 m^3/s *[3 pt; -1 pt if incorrect/missing units]*

65. A 4.03-cm storm hit the Ilex Forest. During this storm, Professor Elm measured throughfall and stemflow on a ⅛-ha plot. Tanks collected 27,600 L of stemflow. The throughfall was collected in rain gauges scattered over the plot, and it averaged 2.49 cm. Calculate the depth and percentage of rainfall intercepted. Round to the nearest hundredth. [11 pt]

27,600 L = 27.6 m^3 [1 pt]
0.2 ha plot = 2000 m^2 [1 pt]
27.6 m^3 / 2000 m^2 = 0.0138 m [1 pt]
S = 1.38 cm [2 pt]
S + T = 1.38 cm + 2.49 cm = 3.87 cm [1 pt]
I = P - (S + T) = 4.03 cm - 3.87 cm = 0.16 cm [3 pt]
I/P = 0.16 cm / 4.03 cm = 0.0397 = 3.97% [2 pt]

66. Team Rocket collected data on the Poliwrath River and observed floods ~~that occurred in the last 40 years~~. Below is a table depicting the maximum stage and discharge values of floods that occurred in this river. Some data points are missing, as 64 floods occurred ~~in the last 40 years~~ on the Poliwrath River.

Rank	Year	Month	Date	Stage (m)	Discharge (m3/s)
1	1948	May	31	11.0	15,200
2	1972	Jun	16	10.1	12,900
3	1950	Jun	20	9.9	12,500
4	1964	Jun	21	9.6	11,600
5	1997	Jun	5	9.5	11,300
6	1955	Jun	29	9.4	11,300
7	1999	Jun	22	9.4	11,000
8	2007	Jun	10	9.3	10,850
9	1974	Jun	22	9.3	10,800
10	2002	Jun	21	9.2	10,600

a. What is the recurrence interval of the flood that occurred on June 21, 1964? [3 pt]

$R_i = (n+1)/r = (59+1)/4 =$ 15 years *[3 pt; -1 pt for incorrect/missing units]*

b. Calculate the probability that a flood of 11,600 m^3/s will occur next year. Leave your answer as a percentage and round to the nearest hundredth. [3 pt]

Pr(discharge = 11,600 m^3/s) = $1/R_i = 1/15 =$ 6.67% *[3 pt]*

c. Based on Team Rocket's study, identify the trend about when floods on the Poliwrath River occur. [3 pt]

Award 3 pt for mentioning that the **strongest** floods occur in early summer months (i.e. May and June). Don't give points for just saying that floods occur in May/June, as there are many data points missing, and the table only shows the 10 strongest floods.
Sample answer: Floods with the highest discharges typically happen in May/June